Data Analytics

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Agenda

Must include the following, but not limited to:

1. About the company
2. Assigned Project details
3. Technologies used
4. Team and role
5. Results and snapshots
6. Conclusion
7. Internship Completion certificate/status

1. About the Company

- **CodTech** is a platform that provides industry-oriented internship programs in various technical domains such as Data Analysis, Machine Learning, and Web Development.

- The organization focuses on practical learning by assigning real-world tasks that help students gain hands-on experience.

- The internship program is designed to enhance technical skills, analytical thinking, and problem-solving abilities of students.

2. Assigned Project

The internship project includes multiple data analysis tasks:

- Exploratory Data Analysis (EDA)
- House Price Prediction (Machine Learning)
- Sales Dashboard Development
- Sentiment Analysis using NLP

Goal:
To understand real-world data handling and build practical solutions

3. Technologies Used

Programming Language:

- Python

Libraries:

- Pandas – data manipulation
- NumPy – numerical operations
- Matplotlib & Seaborn – visualization
- Scikit-learn – machine learning

Tools:

- VS Code
- Jupyter Notebook
- Dash & Plotly (Dashboard)

RESULT
Task1: Big Data Analysis

Description:
Performed exploratory data analysis on dataset

Steps Performed:

- Loaded dataset using Pandas
- Checked data using .info() and .describe()
- Handled missing values
- Created visualizations

Result:
Identified patterns and trends

Outcome:
Learned data preprocessing and visualization

```python
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(x="COUNTRY", y="SALES", data=pdf)
plt.xticks(rotation=90)
plt.show()
```

[10]

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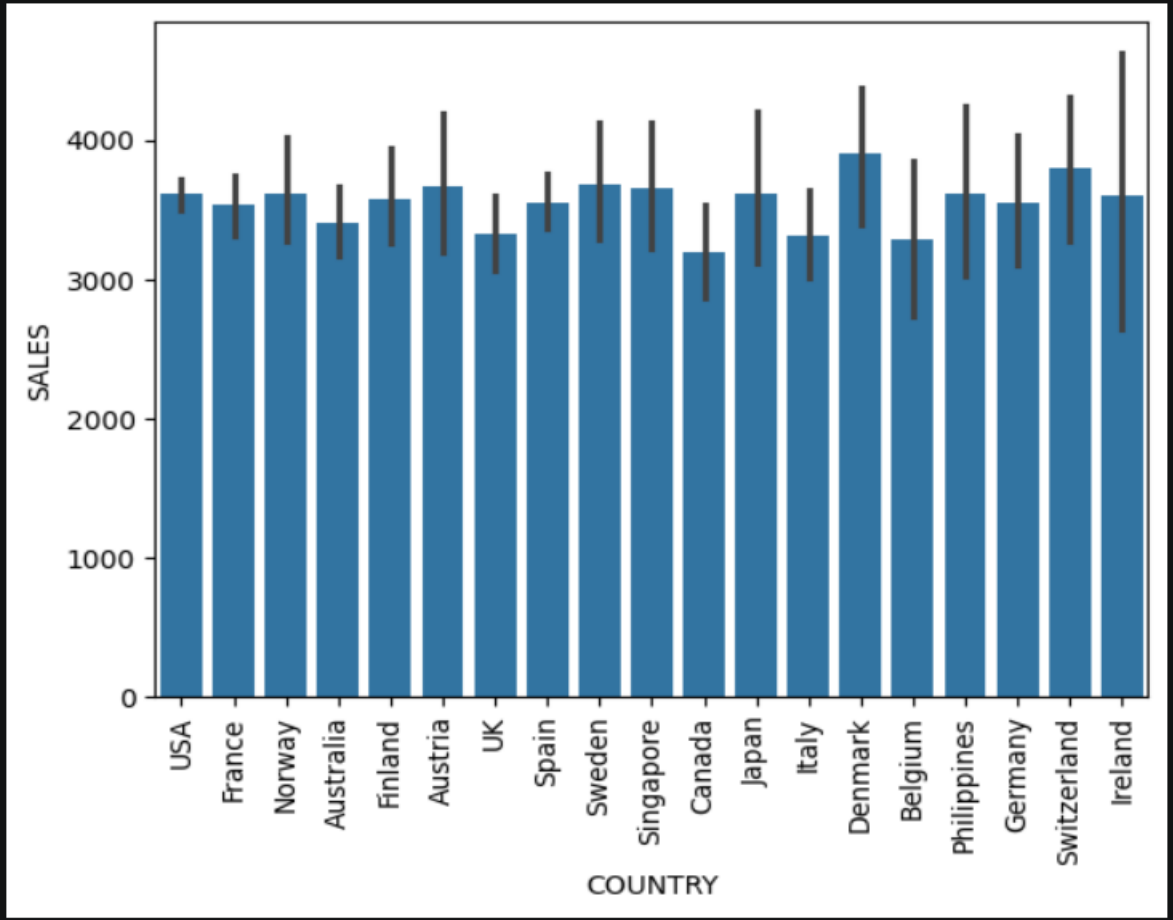

Fig 1: Sales Distribution Across Different Countries

This bar chart shows sales across different countries. It helps compare performance and identify which countries have higher or lower sales.

Task 2: Machine Learning(House Price Prediction)

Description:
Built ML model to predict house prices

Steps Performed:

- Selected features and target variable
- Split dataset (train-test)
- Applied Linear Regression
- Evaluated using MSE & R^2 score

Result:
Model achieved good prediction accuracy

Outcome:
Learned model training and evaluation

```python
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

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```
Mean Squared Error: 24.291119474973538
R2 Score: 0.6687594935356317
```

Fig 2: Model Evaluation Metrics (MSE and R^2 Score)

This output shows the performance of the regression model using Mean Squared Error (MSE) and R^2 score. The MSE indicates the prediction error, while the R^2 score represents how well the model explains the data.

```python
plt.scatter(y_test, predictions)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()
```

[10]

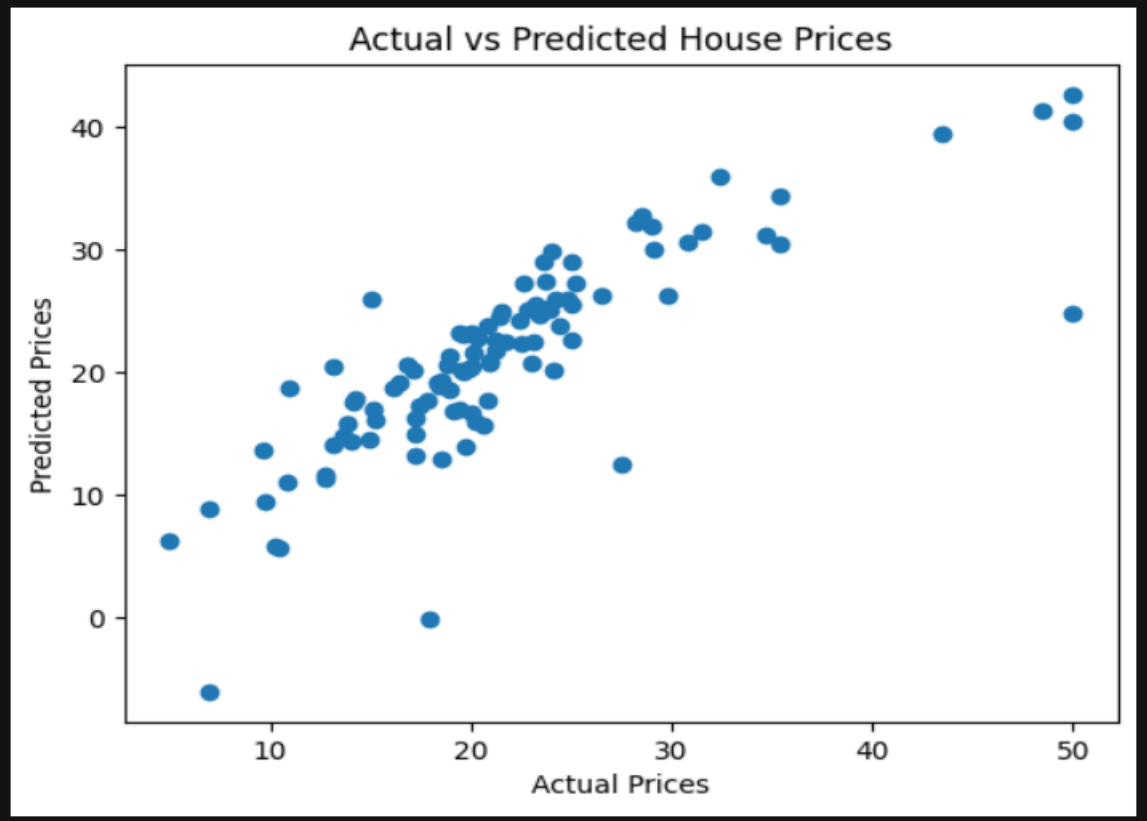

Fig 3: Actual vs Predicted House Prices

This scatter plot compares actual house prices with predicted values from the model. The closer the points are to a straight line, the better the model's predictions, indicating good accuracy.

Task 3: Dashboard Development

Description:
Developed interactive sales dashboard

Steps Performed:

- Loaded dataset
- Created bar, line, and pie charts
- Used Dash & Plotly
- Added dropdown filters

Result:
Interactive dashboard showing insights

Outcome:
Improved data visualization skills

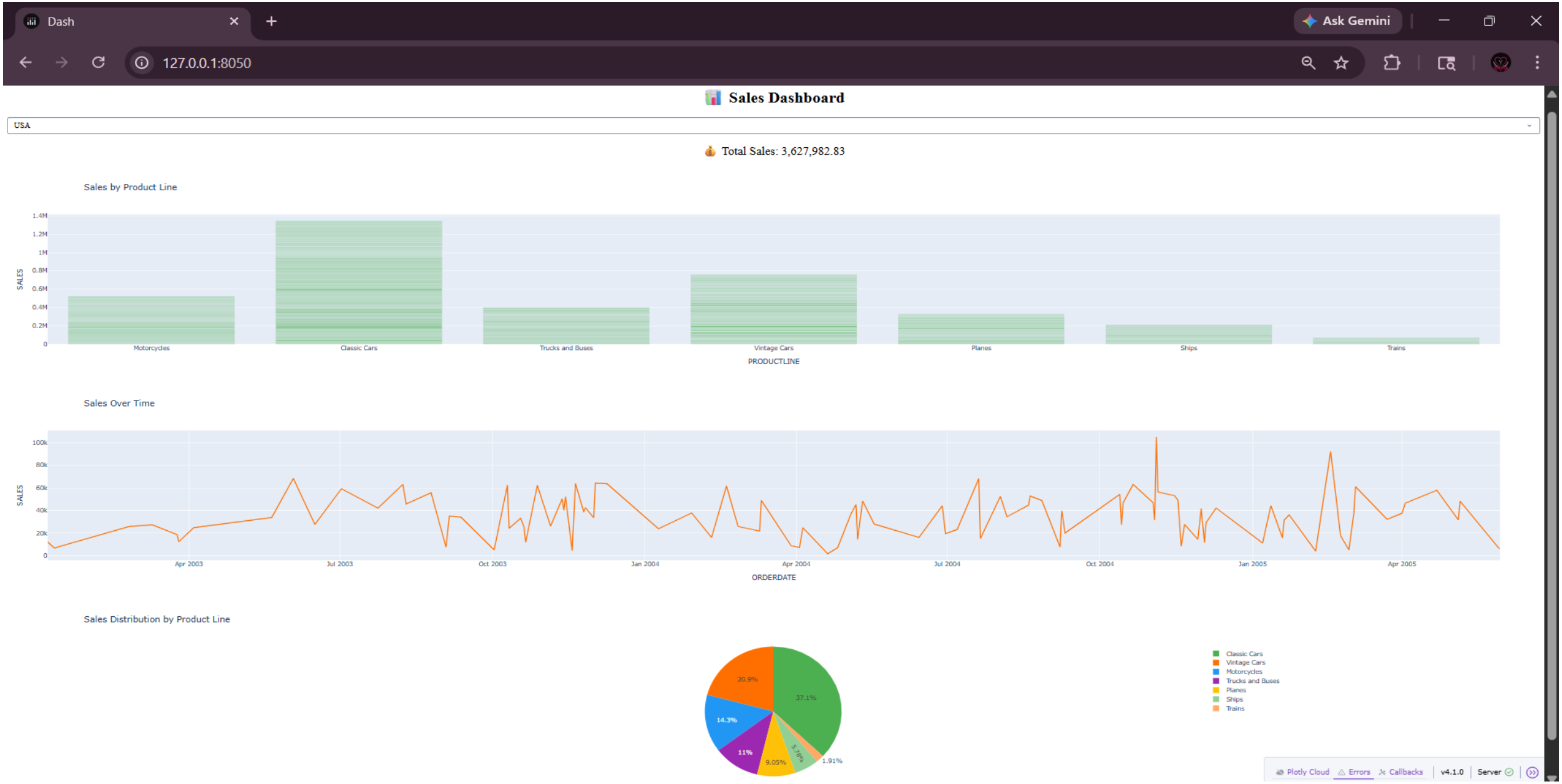

Fig 4: Interactive Sales Dashboard

This dashboard displays sales insights using interactive visualizations, including bar, line, and pie charts. It allows users to analyze sales by product line, track sales over time, and view distribution across categories.

Task 4: Sentiment Analysis

Description:
Classified reviews into positive & negative

Steps Performed:

- Cleaned text data
- Applied TF-IDF Vectorization
- Used Logistic Regression
- Evaluated accuracy (~91%)

Result:
Model successfully classified sentiments

Outcome:
Learned NLP basics and text classification

```python
#MODEL TRAINING
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train_vec, y_train)

#PREDICTION
y_pred = model.predict(X_test_vec)

#EVALUATION
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.91424
```

Fig 5: Logistic Regression Model Training and Accuracy

This output shows the training and evaluation of the Logistic Regression model for sentiment analysis. The model is trained on text data and achieves an accuracy of approximately 91%, indicating good performance in classification.

```python
train_df["Sentiment"].value_counts().plot(
    kind="bar",
    color=colors
)

plt.title("Sentiment Distribution", fontsize=14)
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```

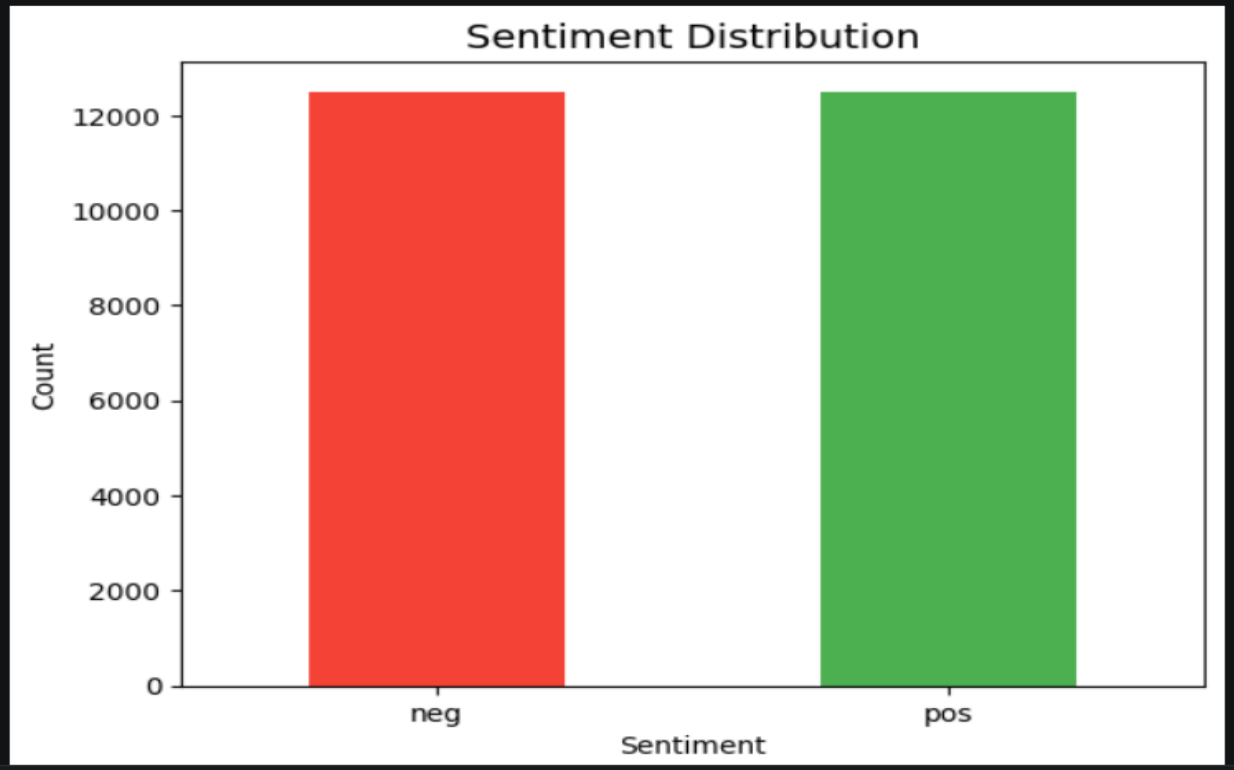

Fig 6: Sentiment Distribution Visualization

This visualization represents the distribution of sentiment labels (positive and negative) in the dataset. The bar chart shows that both sentiment classes are nearly balanced, which helps in reducing bias during model training. A balanced dataset improves the performance and reliability of the sentiment analysis model.

```python
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap="Blues")  # blue theme

plt.title("Confusion Matrix")
plt.show()
```

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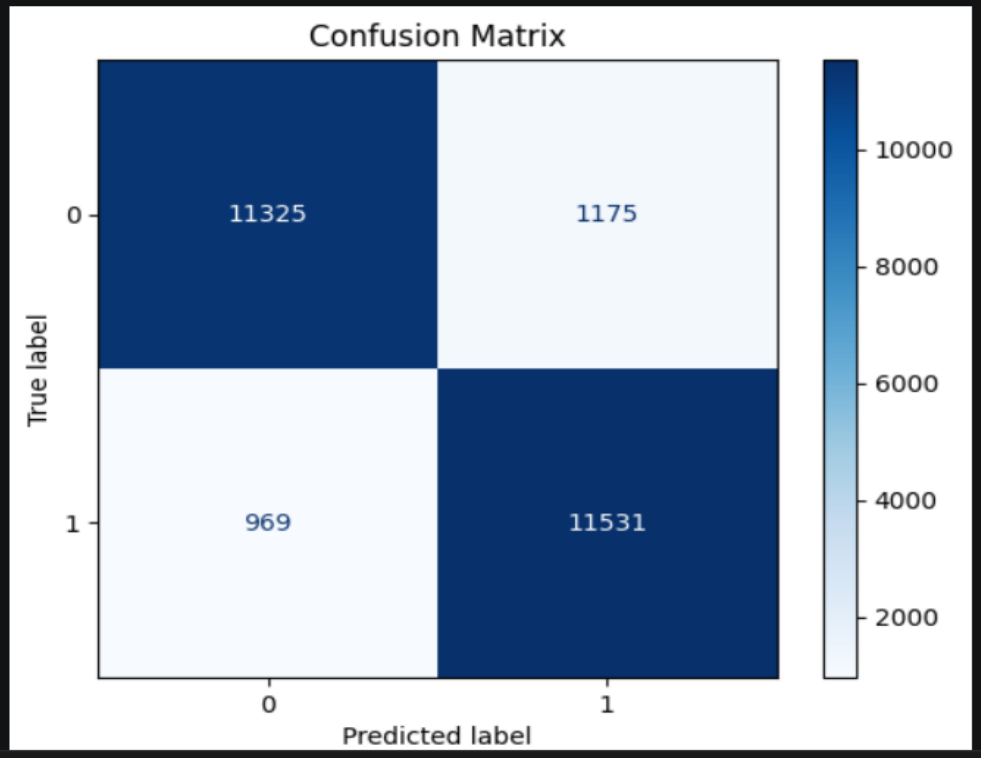

Fig 7: Confusion Matrix

This confusion matrix visualizes the performance of the sentiment analysis model by comparing actual and predicted labels. It shows the number of correct predictions as well as misclassifications, helping to evaluate how well the model distinguishes between positive and negative sentiments.

```python
review = ["This movie was absolutely amazing and I loved it"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

[46]

```
Positive Review 😊
```

```python
review = ["worst movie ever"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

[49]

```
Negative Review 😞
```

Fig 8: Sentiment Prediction on Sample Reviews

This output demonstrates the model predicting sentiment for new text inputs. The model correctly classifies a positive review and a negative review, showing its effectiveness in real-time sentiment prediction.

4. Conclusion

- Completed full data analysis pipeline
- Worked on real-world datasets
- Built ML models and dashboards
- Performed sentiment analysis

Skills gained:

- Python
- Machine Learning
- Data Visualization
- NLP

5. Internship Completion certificate/status

- All 4 tasks completed successfully
- Project submitted
- Certificate: Awaited

Thank you